

Lista 03

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```
library(DBI)
library(RSQLite)

# Conecta/Cria o arquivo
con <- dbConnect(RSQLite::SQLite(), "me315.sqlite")
```

TABELA PRESERVE

```
SELECT * FROM PRESERVE
```

Table 1: Displaying records 1 - 10

PNO	PNAME	STATE	ACRES	FEE
5	HASSAYAMPA RIVER	AZ	660	3
3	DANCING PRAIRIE	MT	680	0
7	MULESHOE RANCH	AZ	49120	0
40	SOUTH FORK MADISON	MT	121	0
14	MCELWAIN-OLSEN	MA	66	0
13	TATKON	MA	40	0
9	DAVID H. SMITH	MA	830	0
11	MIACOMET MOORS	MA	4	0
12	MOUNT PLANTAIN	MA	730	0
1	COMERTOWN PRAIRIE	MT	1130	0

TABELA EMPLOYEE

```
SELECT * FROM EMPLOYEE
```

Table 2: 6 records

ENO	ENAME	SALARY	DNO
1000	MOE	2000	20
2000	LARRY	2000	10
3000	CURLY	3000	20
4000	SHEMP	500	40
5000	JOE	400	10
6000	GEORGE	9000	20

1. Exiba os valores de STATE e FEE da tabela PRESERVE mas sem duplicados.

```
SELECT DISTINCT STATE, FEE  
FROM PRESERVE
```

Table 3: 4 records

STATE	FEE
AZ	3
MT	0
AZ	0
MA	0

2. Exiba todos os valores de DNO na tabela EMPLOYEE. Não exiba valores duplicados.

```
SELECT DISTINCT DNO  
FROM EMPLOYEE
```

Table 4: 3 records

DNO
20
10
40

3. Execute as QUERYs a seguir. Examine as tabelas resultantes e faça as observações relevantes.

```
SELECT DNO, SALARY FROM EMPLOYEE
```

Table 5: 6 records

DNO	SALARY
20	2000
10	2000
20	3000
40	500
10	400
20	9000

```
SELECT DISTINCT DNO, SALARY FROM EMPLOYEE
```

Table 6: 6 records

DNO	SALARY
20	2000
10	2000
20	3000
40	500
10	400
20	9000

```
SELECT DISTINCT DNO, SALARY FROM EMPLOYEE ORDER BY DNO, SALARY
```

Table 7: 6 records

DNO	SALARY
10	400
10	2000
20	2000
20	3000
20	9000
40	500

Primeira query: Mostra todas as 6 linhas da tabela na ordem em que foram salvas, sem nenhum tipo de filtro ou ordenação.

Segunda query: O DISTINCT avalia as duas colunas juntas; como não existe nenhum funcionário com o mesmo departamento e o mesmo salário ao mesmo tempo, nenhuma linha é repetida e o resultado continua trazendo as 6 linhas.

Terceira query: Pega os mesmos resultados únicos e os organiza em ordem crescente pelo departamento e, em caso de empate no departamento, desempatando pelo menor salário.

- Exiba todas as informações sobre qualquer reserva natural em Montana que seja menor que 1.000 acres.

```
SELECT *
FROM PRESERVE
WHERE STATE = 'MT' AND ACRES < 1000
```

Table 8: 2 records

PNO	PNAME	STATE	ACRES	FEE
3	DANCING PRAIRIE	MT	680	0
40	SOUTH FORK MADISON	MT	121	0

- Exiba todas as informações sobre qualquer reserva natural cujo valor de ACRES esteja entre 1200 e 15000, inclusive.

```
SELECT *
FROM PRESERVE
WHERE ACRES >= 1200 AND ACRES <= 15000
```

Table 9: 2 records

PNO	PNAME	STATE	ACRES	FEE
2	PINE BUTTE SWAMP	MT	15000	0
6	PAPAGONIA-SONOITA CREEK	AZ	1200	3

- Exiba todas as informações sobre qualquer reserva natural localizada em Montana que não possua taxa de entrada e seja maior que 10.000 acres.

```
SELECT *
FROM PRESERVE
WHERE FEE = 0 AND ACRES > 10000
```

Table 10: 2 records

PNO	PNAME	STATE	ACRES	FEE
7	MULESHOE RANCH	AZ	49120	0
2	PINE BUTTE SWAMP	MT	15000	0

7. Exiba todas as informações sobre as reservas naturais localizadas em Montana ou Massachusetts.

```
SELECT *
FROM PRESERVE
WHERE STATE = 'MA' OR STATE = 'MT'
```

Table 11: Displaying records 1 - 10

PNO	PNAME	STATE	ACRES	FEE
3	DANCING PRAIRIE	MT	680	0
40	SOUTH FORK MADISON	MT	121	0
14	MCELWAIN-OLSEN	MA	66	0
13	TATKON	MA	40	0
9	DAVID H. SMITH	MA	830	0
11	MIACOMET MOORS	MA	4	0
12	MOUNT PLANTAIN	MA	730	0
1	COMERTOWN PRAIRIE	MT	1130	0
2	PINE BUTTE SWAMP	MT	15000	0
10	HOFT FARM	MA	90	0

8. Exiba todas as informações sobre qualquer reserva natural localizada em Montana ou qualquer reserva que seja menor que 1.000 acres.

```
SELECT *
FROM PRESERVE
WHERE STATE = 'MT' OR ACRES < 1000
```

Table 12: Displaying records 1 - 10

PNO	PNAME	STATE	ACRES	FEE
5	HASSAYAMPA RIVER	AZ	660	3
3	DANCING PRAIRIE	MT	680	0
40	SOUTH FORK MADISON	MT	121	0

PNO	PNAME	STATE	ACRES	FEE
14	MCELWAIN-OLSEN	MA	66	0
13	TATKON	MA	40	0
9	DAVID H. SMITH	MA	830	0
11	MIACOMET MOORS	MA	4	0
12	MOUNT PLANTAIN	MA	730	0
1	COMERTOWN PRAIRIE	MT	1130	0
2	PINE BUTTE SWAMP	MT	15000	0

9. Exiba o número e o nome de qualquer reserva natural que possua taxa de entrada que não seja igual a zero.

```
SELECT PNO, PNAME
FROM PRESERVE
WHERE NOT FEE = 0
```

Table 13: 3 records

PNO	PNAME
5	HASSAYAMPA RIVER
80	RAMSEY CANYON
6	PAPAGONIA-SONOITA CREEK

10. Exiba o número e o nome das reservas naturais que não possuem taxa de entrada de US\$ 3,00 e não possuem taxa de US\$ 10,00.

```
SELECT PNO, PNAME, FEE
FROM PRESERVE
WHERE NOT (FEE = 3 OR FEE = 10)
```

Table 14: Displaying records 1 - 10

PNO	PNAME	FEE
3	DANCING PRAIRIE	0
7	MULESHOE RANCH	0
40	SOUTH FORK MADISON	0
14	MCELWAIN-OLSEN	0
13	TATKON	0
9	DAVID H. SMITH	0

PNO	PNAME	FEE
11	MIACOMET MOORS	0
12	MOUNT PLANTAIN	0
1	COMERTOWN PRAIRIE	0
2	PINE BUTTE SWAMP	0

11. Exiba todas as informações sobre qualquer reserva natural localizada no Arizona que não possua taxa de entrada, ou qualquer reserva que seja menor que 100 acres (independentemente de seus valores de STATE e FEE).

```
SELECT *
FROM PRESERVE
WHERE (STATE = 'AZ' AND FEE = 0) OR ACRES < 100
```

Table 15: 5 records

PNO	PNAME	STATE	ACRES	FEE
7	MULESHOE RANCH	AZ	49120	0
14	MCELWAIN-OLSEN	MA	66	0
13	TATKON	MA	40	0
11	MIACOMET MOORS	MA	4	0
10	HOFT FARM	MA	90	0

12. Exiba todas as informações sobre qualquer reserva natural que seja menor que 1.000 acres e tenha taxa de entrada de zero dólares ou esteja localizada no Arizona.

```
SELECT *
FROM PRESERVE
WHERE (ACRES < 1000 AND FEE = 0) OR STATE = 'AZ'
```

Table 16: Displaying records 1 - 10

PNO	PNAME	STATE	ACRES	FEE
5	HASSAYAMPA RIVER	AZ	660	3
3	DANCING PRAIRIE	MT	680	0
7	MULESHOE RANCH	AZ	49120	0
40	SOUTH FORK MADISON	MT	121	0
14	MCELWAIN-OLSEN	MA	66	0
13	TATKON	MA	40	0

PNO	PNAME	STATE	ACRES	FEE
9	DAVID H. SMITH	MA	830	0
11	MIACOMET MOORS	MA	4	0
12	MOUNT PLANTAIN	MA	730	0
80	RAMSEY CANYON	AZ	380	3

13. Selecione todas as informações sobre qualquer reserva natural com taxa de entrada que não seja maior que zero, ou qualquer outra reserva, independentemente de sua taxa, que esteja localizada em Montana e seja maior que 1.000 acres.

```
SELECT *
FROM PRESERVE
WHERE (NOT FEE > 0) OR (STATE = 'MT' AND ACRES > 1000)
```

Table 17: Displaying records 1 - 10

PNO	PNAME	STATE	ACRES	FEE
3	DANCING PRAIRIE	MT	680	0
7	MULESHOE RANCH	AZ	49120	0
40	SOUTH FORK MADISON	MT	121	0
14	MCELWAIN-OLSEN	MA	66	0
13	TATKON	MA	40	0
9	DAVID H. SMITH	MA	830	0
11	MIACOMET MOORS	MA	4	0
12	MOUNT PLANTAIN	MA	730	0
1	COMERTOWN PRAIRIE	MT	1130	0
2	PINE BUTTE SWAMP	MT	15000	0

14. Exiba todas as informações sobre todas as reservas naturais, exceto aquelas reservas de Montana sem taxa de entrada.

```
SELECT *
FROM PRESERVE
WHERE NOT (STATE = 'MT' OR FEE = 0)
```

Table 18: 3 records

PNO	PNAME	STATE	ACRES	FEE
5	HASSAYAMPA RIVER	AZ	660	3

PNO	PNAME	STATE	ACRES	FEE
80	RAMSEY CANYON	AZ	380	3
6	PAPAGONIA-SONOITA CREEK	AZ	1200	3

15. As seguintes cláusulas WHERE são logicamente equivalentes?

```
SELECT *
FROM PRESERVE
WHERE STATE = 'MA' AND (ACRES > 1000 OR FEE = 0.0)
```

Table 19: 6 records

PNO	PNAME	STATE	ACRES	FEE
14	MCELWAIN-OLSEN	MA	66	0
13	TATKON	MA	40	0
9	DAVID H. SMITH	MA	830	0
11	MIACOMET MOORS	MA	4	0
12	MOUNT PLANTAIN	MA	730	0
10	HOFT FARM	MA	90	0

```
SELECT *
FROM PRESERVE
WHERE (STATE = 'MA' AND ACRES > 1000) OR (STATE = 'MA' AND FEE = 0.0)
```

Table 20: 6 records

PNO	PNAME	STATE	ACRES	FEE
14	MCELWAIN-OLSEN	MA	66	0
13	TATKON	MA	40	0
9	DAVID H. SMITH	MA	830	0
11	MIACOMET MOORS	MA	4	0
12	MOUNT PLANTAIN	MA	730	0
10	HOFT FARM	MA	90	0

Resposta: Elas são logicamente equivalentes, pois $C1 \text{ AND } (C2 \text{ OR } C3) = (C1 \text{ AND } C2) \text{ OR } (C1 \text{ AND } C3)$.

16. As seguintes cláusulas WHERE são logicamente equivalentes?

```
SELECT *
FROM PRESERVE
WHERE STATE = 'MA' OR (ACRES > 1000 AND FEE = 0.0)
```

Table 21: 9 records

PNO	PNAME	STATE	ACRES	FEE
7	MULESHOE RANCH	AZ	49120	0
14	MCELWAIN-OLSEN	MA	66	0
13	TATKON	MA	40	0
9	DAVID H. SMITH	MA	830	0
11	MIACOMET MOORS	MA	4	0
12	MOUNT PLANTAIN	MA	730	0
1	COMERTOWN PRAIRIE	MT	1130	0
2	PINE BUTTE SWAMP	MT	15000	0
10	HOFT FARM	MA	90	0

```
SELECT *
FROM PRESERVE
WHERE (STATE = 'MA' OR ACRES > 1000) AND (STATE = 'MA' OR FEE = 0.0)
```

Table 22: 9 records

PNO	PNAME	STATE	ACRES	FEE
7	MULESHOE RANCH	AZ	49120	0
14	MCELWAIN-OLSEN	MA	66	0
13	TATKON	MA	40	0
9	DAVID H. SMITH	MA	830	0
11	MIACOMET MOORS	MA	4	0
12	MOUNT PLANTAIN	MA	730	0
1	COMERTOWN PRAIRIE	MT	1130	0
2	PINE BUTTE SWAMP	MT	15000	0
10	HOFT FARM	MA	90	0

Resposta: Elas são logicamente equivalentes, pois $C1 \text{ OR } (C2 \text{ AND } C3) = (C1 \text{ OR } C2) \text{ AND } (C1 \text{ OR } C3)$.

17. As seguintes cláusulas WHERE são logicamente equivalentes?

```
SELECT *
FROM PRESERVE
WHERE NOT (ACRES < 50 AND STATE = 'MA')
```

Table 23: Displaying records 1 - 10

PNO	PNAME	STATE	ACRES	FEE
5	HASSAYAMPA RIVER	AZ	660	3
3	DANCING PRAIRIE	MT	680	0
7	MULESHOE RANCH	AZ	49120	0
40	SOUTH FORK MADISON	MT	121	0
14	MCELWAIN-OLSEN	MA	66	0
9	DAVID H. SMITH	MA	830	0
12	MOUNT PLANTAIN	MA	730	0
1	COMERTOWN PRAIRIE	MT	1130	0
2	PINE BUTTE SWAMP	MT	15000	0
80	RAMSEY CANYON	AZ	380	3

```
SELECT *
FROM PRESERVE
WHERE NOT ACRES < 50 AND NOT STATE = 'MA'
```

Table 24: 8 records

PNO	PNAME	STATE	ACRES	FEE
5	HASSAYAMPA RIVER	AZ	660	3
3	DANCING PRAIRIE	MT	680	0
7	MULESHOE RANCH	AZ	49120	0
40	SOUTH FORK MADISON	MT	121	0
1	COMERTOWN PRAIRIE	MT	1130	0
2	PINE BUTTE SWAMP	MT	15000	0
80	RAMSEY CANYON	AZ	380	3
6	PAPAGONIA-SONOITA CREEK	AZ	1200	3

```
SELECT *
FROM PRESERVE
WHERE ACRES >= 50 OR STATE <> 'MA'
```

Table 25: Displaying records 1 - 10

PNO	PNAME	STATE	ACRES	FEE
5	HASSAYAMPA RIVER	AZ	660	3
3	DANCING PRAIRIE	MT	680	0
7	MULESHOE RANCH	AZ	49120	0
40	SOUTH FORK MADISON	MT	121	0
14	MCELWAIN-OLSEN	MA	66	0
9	DAVID H. SMITH	MA	830	0
12	MOUNT PLANTAIN	MA	730	0
1	COMERTOWN PRAIRIE	MT	1130	0
2	PINE BUTTE SWAMP	MT	15000	0
80	RAMSEY CANYON	AZ	380	3

Resposta: Somente a primeira e a terceira cláusulas WHERE são equivalentes, pois NOT (C1 AND C2) = (NOT C1) OR (NOT C2). Para a segunda ser equivalente ela deveria ser WHERE NOT ACRES < 50 OR STATE <> 'MA'.

18. As seguintes cláusulas WHERE são logicamente equivalentes?

```
SELECT *
FROM PRESERVE
WHERE NOT (ACRES < 50 OR STATE = 'MA')
```

Table 26: 8 records

PNO	PNAME	STATE	ACRES	FEE
5	HASSAYAMPA RIVER	AZ	660	3
3	DANCING PRAIRIE	MT	680	0
7	MULESHOE RANCH	AZ	49120	0
40	SOUTH FORK MADISON	MT	121	0
1	COMERTOWN PRAIRIE	MT	1130	0
2	PINE BUTTE SWAMP	MT	15000	0
80	RAMSEY CANYON	AZ	380	3
6	PAPAGONIA-SONOITA CREEK	AZ	1200	3

```
SELECT *
FROM PRESERVE
WHERE NOT ACRES < 50 OR NOT STATE = 'MA'
```

Table 27: Displaying records 1 - 10

PNO	PNAME	STATE	ACRES	FEE
5	HASSAYAMPA RIVER	AZ	660	3
3	DANCING PRAIRIE	MT	680	0
7	MULESHOE RANCH	AZ	49120	0
40	SOUTH FORK MADISON	MT	121	0
14	MCELWAIN-OLSEN	MA	66	0
9	DAVID H. SMITH	MA	830	0
12	MOUNT PLANTAIN	MA	730	0
1	COMERTOWN PRAIRIE	MT	1130	0
2	PINE BUTTE SWAMP	MT	15000	0
80	RAMSEY CANYON	AZ	380	3

```
SELECT *
FROM PRESERVE
WHERE NOT ACRES < 50 AND NOT STATE = 'MA'
```

Table 28: 8 records

PNO	PNAME	STATE	ACRES	FEE
5	HASSAYAMPA RIVER	AZ	660	3
3	DANCING PRAIRIE	MT	680	0
7	MULESHOE RANCH	AZ	49120	0
40	SOUTH FORK MADISON	MT	121	0
1	COMERTOWN PRAIRIE	MT	1130	0
2	PINE BUTTE SWAMP	MT	15000	0
80	RAMSEY CANYON	AZ	380	3
6	PAPAGONIA-SONOITA CREEK	AZ	1200	3

Resposta: Somente a primeira e a terceira cláusulas WHERE são equivalentes, pois NOT (C1 OR C2) = (NOT C1) AND (NOT C2). Para a segunda ser equivalente ela deveria ser igual a terceira.

```
Sys.setenv(TZ = "America/Sao_Paulo")
Sys.time()
```

```
[1] "2026-09-01 00:03:49 -03"
```